

ESRP-CMS-TPCB300

ThinManager Ready Modular TPC - Computing Box Module with Intel® Elkhart Lake



Features

- Computing Box powered by Intel® Atom™ X6425E with DDR4 SODIMM slot (up to 32GB)
- Verified with ThinManager, ThinManager BIOS enabled
- Multiple sized front panel module offering with P-Cap multi-touch sensor
- 1 x M.2 Key-E 2230 for WIFI + BT module
- 2 x 2.5 Gb LAN
- Supports TPM2.0 hardware security
- Wide Temp. Support. Range -20~60 °C (Atom™ X6425E)

Introduction

Advantech ThinManager compatible industrial thin clients exclusively integrate with ThinManager® software and are verified for ThinManager distributed applications. The thin clients are used to display content delivered from a remote ThinManager server. Multiple thin clients set up across the plant or facility with various roles and all maintained by the ThinManager server. All content remains on the server so downtime is reduced in the event of a failure. ESRP-CMS-TPCB300 is Small-Size Embedded Thin Client with Intel® Atom X6425E. ESRP-CMS-TPCB300 allows users to change the panel sizes (12"-23.8").

Specifications

General

- **Certification** CE, FCC Class A, CB/UL
- **Cooling System** Fanless design
- **Dimensions (W x H x D)** 269 x 203 x 30 (mm)
- **Enclosure** Metal
- **Mounting** *VESA Mount, Wall or Panel Mount
*VESA mount required TPC-5000T-VMKE
- **OS Support** Windows 10 IoT Enterprise LTSC
- **Power Input** 24 V_{DC} ± 20%
- **Weight (Net)** 2.1 kg (4.6 lbs)

System Hardware

- **CPU** Intel® Elkhart Lake
Atom™ x6425E 2.00 GHz, quad core
- **Memory** 1 x DDR4 SO-DIMM slot (up to 32GB)
- **Storage** 1 x M.2 M key 2280 (SATA/PCIe)
1 x 2.5" SSD/HDD
- **LAN** 2 x Intel® I226; 2.5GbE
- **Expansion Slot** 1 x Full-size mini PCIe (PCIe/USB2.0)
*Support iDoor module / 4G&5G LTE SIM slot
1 x M.2 AE key 2230 (PCIe/USB2.0)
- **I/O** 2 x RS-232/422/485
3 x USB 3.2
1 x USB 2.0
1 x Line out
1 x Display Port (4096x2160@60Hz)
- **Power Input** 24 V_{DC} ± 20%
- **Security** TPM 2.0 (built-in)

Environment

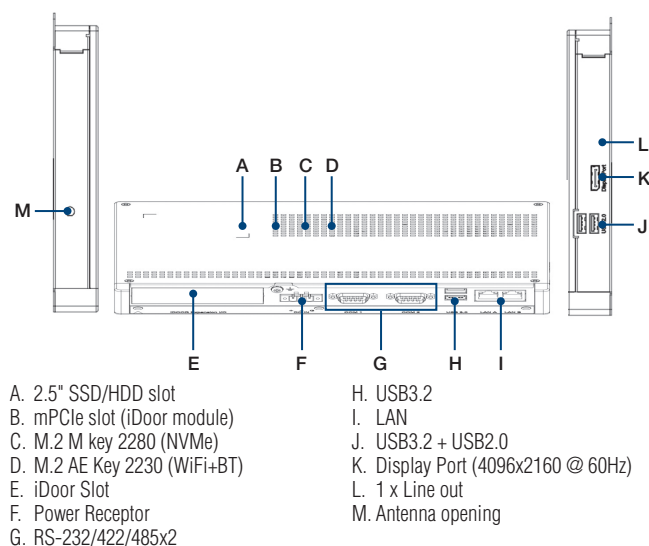
- **Humidity** 95% RH @ 40°C, non-condensing
- **Ingress Protection** IP66 (for optional panel module with "Panel Mount")
- **Operating Temperature** -20°C~60°C
- **Storage Temperature** -20°C~70°C

Power Consumption with Panel Module

Item	Power Consumption		Screen Size	Resolution	Aspect Ratio
	Typ.	Max.			
TPC-B300	21W	50W	N/A	N/A	N/A
w/ FPM-D12T-CE	33W	62W	12.1"	1024x768	04:03
w/ FPM-D12T-HE	35W	64W	12.1"	1024x768	
w/ FPM-D15T-CE	30W	59W	15"	1024x768	
w/ FPM-D15T-HE	45W	74W	15"	1024x768	
w/ FPM-D17T-CE	37W	66W	17"	1280x1024	05:04
w/ FPM-D19T-CE	35W	64W	19"	1280x1024	
w/ FPM-D15W-FCE	37W	66W	15.6"	1920x1080	16:09
w/ FPM-D15W-FHE	46W	75W	15.6"	1920x1080	
w/ FPM-D18W-FCE	38W	67W	18.5"	1920x1080	
w/ FPM-D21W-CE	33W	62W	21.5"	1920x1080	
w/ FPM-D21W-HE	63W	92W	21.5"	1920x1080	
w/ FPM-D24W-HE	42W	71W	23.8"	1920x1080	

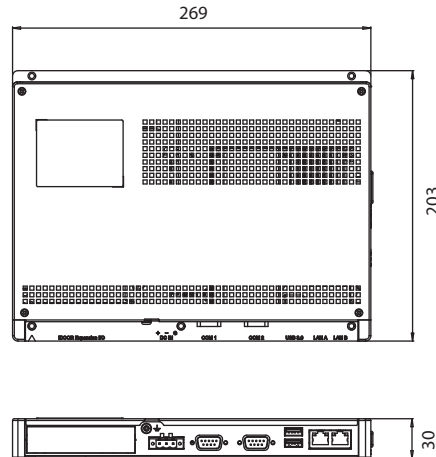
*Please find below the Ordering Information for the Panel Module.

Rear View



Dimensions

Unit: mm



Ordering Information

- **ESRP-CMS-TPCB300** Thin client with TPC-B300

Panel Module P/N	FPM-D12T-CE	FPM-D12T-HE	FPM-D15T-CE	FPM-D15T-HE	FPM-D17T-CE	FPM-D19T-CE
Description	12.1" XGA TFT LED LCD	12.1" XGA TFT LED LCD High Brightness 1000 nits	15" XGA TFT LED LCD	15" XGA TFT LED LCD High Brightness 1800 nits	17" SXGA TFT LED LCD	19" SXGA TFT LED LCD

Panel Module P/N	FPM-D15W-FCE	FPM-D15W-FHE	FPM-D18W-FCE	FPM-D21W-CE	FPM-D21W-HE	FPM-D24W-CE
Description	15.6" FHD TFT LED LCD	15.6" FHD TFT LED LCD High Brightness 1200 nits	18.5" HD TFT LED LCD	18.5" FHD TFT LED LCD	18.5" FHD TFT LED LCD High brightness 1400 nits	23.8" FHD TFT LED LCD

Note: Display modules cannot work independently. A computing box (TPC-B200/300/510/600) is compulsory to operate the display module.

Packing List

Item	Part No.	Depiction	Qty
Grounding cable	1703011001-11	A Cable G-TEM 4.3 100CM C=GR/Y	1
Plug-In Block	1652007930-01	Plug-In Block 1x3P/5.08mm/PA66/F/VA	1
Warranty	2190000902	CARDBOARD-WARRANTY REV. A2	1
BOX Fix Screw	1930000686	Screw M3x5L S/S D=5 H=1.3 (2+) ST Ni NK	5
iDoor Screw	1930002161	Screw M3x4.0(10.0)L R/S D=5.3 H=2 2+ ST BZn	2
HDD Screw	1930001361	Screw M3x4L S/S D=5 H=1.2 (2+) ST Ni NK	4
CPU Thermal Pad (Celeron J6412)	1990031130N010	Thermal Pad 25x14.7x0.5 K=13 TP PG80B	1
CPU Thermal Pad (Atom X6425E)	1990040910N000	Thermal pad for TPC-B300 PG80B 32X21X0.5 K=13	1
DDR Thermal Pad	1990021846N000	Thermal pad 59x19.5x1.0 K=6 GR-Hm SPC-1840WP	1
NVMe Thermal Pad	1990008165N010	Thermal pad 64x21.5x1.5mm K=2.5 GR25A-OHUNO-3282	1
M.2 Key-E Thermal Pad	1990035990N000	Thermal pad 22x22x1.5mm K=2.5 TP eApus-PG25A	1

Note: Any NVMe SSD with a built-in heatsink may cause interference with the TPC-B300's mechanism and thermal design. Therefore, please purchase an NVMe SSD without a built-in heatsink

Optional Accessories

Part Number	Description
AIW-166K3	6E AX210 (no.vpro), -40~85degree wireless kit
96PSA-A120W24T2-4	ADP A/D 100-240V 120W 24V C14 CORD END 62368
96PSA-A150W24T2-4	A/D 100-240V 150W 24V W/PFC C14 TERMINAL 62368
TPC-5000T-VMKE	TPC Series VESA mount kit for FPM-D Series
1702002600 Power	cable (US) 1.8 M
1702002605 Power	cable (EU) 1.8 M
1702031801 Power	cable (UK) 1.8 M
1700000596 Power	cable (CN/AU) 1.8 M